

A COMPUTATIONAL APPROACH TO THE ENUMERATION OF MAXIMAL G -PARKING FUNCTIONS, REVISITED

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ABSTRACT. Given a connected multigraph G with root q , we give an algorithm that lists every maximal G -parking function exactly once, with no duplicate suppression, by enumerating the lexicographically minimal Dhar burn orders. Its correctness rests on a structure theorem of independent interest: the fibers of the map assigning to each G -parking function its lex-minimal burn order are coordinate boxes which partition the set of all G -parking functions, with box tops exactly the maximal parking functions. One search pass therefore also computes the number of spanning trees $T_G(1, 1)$ and the full degree enumerator $T_G(1, y)$. Our implementation enumerates the 3040575 maximal parking functions of the hypercube Q_4 in under two seconds on a workstation, confirming by direct enumeration the value previously known only through the chromatic polynomial, and computes $T_{Q_4}(1, y)$, which appears to be new. We record the resulting data, including the 133-element catalog for Q_3 that motivated this project.

1. INTRODUCTION

Let $G = (V, E)$ be a connected multigraph with a root $q \in V$. A G -parking function [11] (equivalently a *superstable configuration* of the abelian sandpile model) is a function $f: V \setminus \{q\} \rightarrow \mathbb{Z}_{\geq 0}$ such that every nonempty $A \subseteq V \setminus \{q\}$ contains a vertex v with $f(v) < \sum_{u \notin A} a(v, u)$, where $a(v, u)$ is the edge multiplicity. The number of G -parking functions equals the number of spanning trees of G , and many bijections realizing this are known [4, 6, 2]. The *maximal* G -parking functions (under the coordinatewise order) all have the same total degree $g = |E| - |V| + 1$, are in bijection with the acyclic orientations of G with unique source at q , and number $T_G(1, 0)$, the Tutte evaluation [2].

The original (2016) version of this project sought an algorithm to list the maximal Q_n -parking functions in order to study, experimentally, the formula for the number of spanning trees of the hypercube,

$$(1) \quad c(Q_n) = 2^{2^n - n - 1} \prod_{k=1}^n k^{\binom{n}{k}},$$

for which Stanley's *Enumerative Combinatorics* volume 2 states, at the end of Example 5.6.10, that “a direct combinatorial proof of this formula is not known” [8]. Two combinatorial (though not bijective) proofs were subsequently given by Bernardi [3]; the second edition of [8] (2024, Notes to Chapter 5) records that a *bijective* proof remains unknown. The present update replaces the original breadth-first search over all burn orders — whose running time grew roughly like n^n and which required cluster hardware and a trie to deduplicate output already for Q_4 — by a canonical-order search that produces each maximal parking function exactly once.

Our contributions are:

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- (1) an enumeration algorithm for maximal G -parking functions with no duplicate suppression and $O(|V|)$ working memory per search path (Section 3); we are not aware of any published enumeration algorithm stated for maximal parking functions (poly-delay enumeration of acyclic orientations with a prescribed unique source [5] yields one implicitly through [2, Thm. 3.1]);
- (2) the correctness proof via a fiber theorem: the search leaves are boxes partitioning all of $\mathcal{P}(G, q)$ (Theorem 3.1), which simultaneously yields $T_G(1, 1)$ and $T_G(1, y)$ in the same pass (Corollary 3.2) and constitutes an explicit form of Merino's M -shellability theorem for cographic matroids [10] — see the companion note in this repository for that discussion;
- (3) computational results (Section 4): direct-enumeration confirmation of $T_{Q_4}(1, 0) = 3\,040\,575$ and $c(Q_4) = 42\,467\,328$; the degree enumerator $T_{Q_4}(1, y)$; the catalog for Q_3 ; and sandpile-group data through Q_5 .

2. PRELIMINARIES

Dhar's burning criterion [7]: f is a G -parking function iff, starting from q burnt and repeatedly burning any vertex v whose burnt neighbors have total edge weight exceeding $f(v)$, every vertex burns. The process is abelian, so the following is well defined.

Definition 2.1. The *canonical burn order* $\sigma(f)$ of $f \in \mathcal{P}(G, q)$ is the burn order obtained by always burning the smallest-labeled burnable vertex, for a fixed vertex labeling $q = 0, 1, \dots, n - 1$.

For an ordering σ starting at q , write $w_k(v)$ for the weight of edges from v to $\{\sigma_0, \dots, \sigma_{k-1}\}$, and for $v = \sigma_k$ set $\text{top}_\sigma(v) = w_k(v) - 1$ and $\text{bot}_\sigma(v) = \max\{w_j(v) : j < k, \sigma_j > v\}$ (zero if no such j): the *skip watermark*, i.e. the burnt-neighbor weight of v the last time a larger vertex was burned in preference to it.

3. THE FIBER THEOREM AND THE ALGORITHM

Theorem 3.1. For $f \in \mathcal{P}(G, q)$ with $\sigma = \sigma(f)$,

$$\{g \in \mathcal{P}(G, q) : \sigma(g) = \sigma\} = \{g : \text{bot}_\sigma \leq g \leq \text{top}_\sigma\},$$

and every vector in the box is a parking function. Consequently the boxes partition $\mathcal{P}(G, q)$; the box tops are exactly the maximal parking functions, each arising once.

Proof. If $\sigma(g) = \sigma$ then at step k burnability gives $g(\sigma_k) \leq \text{top}_\sigma(\sigma_k)$, and lex-minimality forces every smaller unburnt u to be unburnable at that step, i.e. $g(u) \geq w_k(u)$; maximizing over the skip steps of u gives $g \geq \text{bot}_\sigma$. Conversely, if $\text{bot}_\sigma \leq g \leq \text{top}_\sigma$, induction on k shows the canonical process for g reproduces σ : σ_k is burnable since $g(\sigma_k) < w_k(\sigma_k)$, and any smaller unburnt u satisfies $g(u) \geq \text{bot}_\sigma(u) \geq w_k(u)$ because u is skipped at step k . Hence all of V burns: $g \in \mathcal{P}(G, q)$ with $\sigma(g) = \sigma$. For the second statement: every edge contributes to $w_k(\sigma_k)$ exactly once over the run, so $\sum_v \text{top}_\sigma(v) = |E| - |V| + 1 = g$, making each top maximal [2, Prop. 2.3, Cor. 3.2]; the top lies in its own box, so $\sigma \mapsto \text{top}_\sigma$ is injective, and a maximal m satisfies $m \leq \text{top}_{\sigma(m)}$ with equal degree, hence equality. $\square$

Corollary 3.2. With $b_m := \text{bot}_{\sigma(m)}$ and $[k]_y = 1 + y + \dots + y^{k-1}$,

$$T_G(1, y) = \sum_{m \text{ maximal}} \prod_{v \neq q} [m(v) - b_m(v) + 1]_y, \quad T_G(1, 1) = \sum_m \prod_{v \neq q} (m(v) - b_m(v) + 1).$$

Proof. Group Merino's identity $T_G(1, y) = \sum_f y^{g - \deg f}$ [9] by the boxes. $\square$

The search. The algorithm is a depth-first search over burn prefixes. The state keeps the burnt set, each vertex's current burnt-neighbor weight $w_{\text{cnt}}(v)$, and its watermark $w_2(v)$. From a prefix, vertex v may be burned iff it is unburnt and $w_{\text{cnt}}(v) > w_2(v)$; burning v sets $w_2(u) \leftarrow w_{\text{cnt}}(u)$ for every smaller unburnt u (their skip watermark) and adds v 's edge weights to its unburnt neighbors. A leaf at depth n emits the box $[\text{bot}, \text{top}]$; in particular its top is one maximal parking function, and by Theorem 3.1 each maximal parking function is emitted exactly once. A cheap dead-end test (an unburnt vertex with $w_{\text{cnt}} = w_2$ and no unburnt neighbor can never burn) prunes hopeless branches. Undoing a burn restores $O(\deg v)$ words, so the working memory per path is $O(|V|)$, the leaves need no deduplication structure, and disjoint subtrees parallelize trivially. This replaces both the n^n -time Python search (2016) and the MPI/trie C implementation (2019) of the earlier drafts.

Correctness and the partition property are validated in the implementation by exhaustive comparison with brute-force Dhar enumeration on Q_1 – Q_3 , on the 4-vertex multigraph example of the 2016 code (reproducing its four maximal parking functions), and on dozens of seeded random connected multigraphs; tree counts are cross-checked against Matrix-Tree determinants and, for hypercubes, against (1).

4. DATA FOR HYPERCUBES

n	g	maximal PFs = $T(1, 0)$	spanning trees	time
2	1	3	4	—
3	5	133	384	ms
4	17	3 040 575	42 467 328	1.5 s
5	49	81 768 640 551 939 777	$\approx 2.08 \times 10^{19}$	<i>not listable</i>

The Q_4 row is, to our knowledge, the first direct-enumeration confirmation of the count 3 040 575, previously derived from the chromatic polynomial of Q_4 ; the Q_5 count is from the chromatic polynomial of Q_5 (OEIS A334278) via $T(1, 0) = |[x] \chi_{Q_n}(x)|$, and a Knuth random-descent estimator over 2×10^8 samples of our search tree gives $(7.4 \pm 0.5) \times 10^{16}$, consistent with it. The degree enumerator of Q_4 -parking functions (equivalently $T_{Q_4}(1, y)$, the h -vector of the cographic matroid of Q_4), computed in the same 1.5-second pass:

1, 15, 120, 680, 3044, 11388, 36808, 104984, 267894, 616906,

1287688, 2436504, 4156516, 6306364, 8278008, 8903016, 7016817, 3040575

(degrees 0 through 17). For Q_3 the row is 1, 7, 28, 76, 139, 133. The Smith normal forms of the reduced Laplacians give the sandpile groups $K(Q_3) = \mathbb{Z}_2 \times \mathbb{Z}_8 \times \mathbb{Z}_{24}$, $K(Q_4) = \mathbb{Z}_2^2 \times \mathbb{Z}_8 \times \mathbb{Z}_{24}^3 \times \mathbb{Z}_{96}$, and $K(Q_5) = \mathbb{Z}_2^5 \times \mathbb{Z}_6 \times \mathbb{Z}_{24}^4 \times \mathbb{Z}_{48} \times \mathbb{Z}_{192}^3 \times \mathbb{Z}_{960}$, consistent with all three theorems of Bai [1]. The full Q_3 catalog (133 maximal parking functions, with their boxes) accompanies the implementation.

5. REMARKS AND QUESTIONS

The 2016 draft asked about the intersections of the down-sets $\text{dom}(f)$ of maximal parking functions, following [2, Cor. 3.3]. Theorem 3.1 answers a sharpened form of that question: the down-sets can be replaced by canonical disjoint boxes, eliminating inclusion–exclusion altogether. Questions we consider open and interesting:

- (1) Is the box partition, in lexicographic order of the burn orders, an M -shelling in the sense of Chari? (Verified computationally on all our test graphs; Merino [10] proved existence inductively.)

- (2) The bottom vector b_m is a statistic on acyclic orientations with unique source that appears to be new; what does it correspond to under the known bijections to safe spanning trees?
- (3) The bijective proof of (1) through this machinery — see the companion paper on the support-product theorem.

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